

Angles, polygons, congruence and vectors - Set 1 - Solutions

Specification M5.1-M5.6, M5.10, M5.14, M5.18, M5.19.

Answer key

Question	Answer	Topic
1	F	Angles, polygons, congruence and vectors
2	E	Angles, polygons, congruence and vectors
3	F	Angles, polygons, congruence and vectors
4	D	Angles, polygons, congruence and vectors
5	C	Angles, polygons, congruence and vectors
6	A	Angles, polygons, congruence and vectors
7	E	Angles, polygons, congruence and vectors
8	F	Angles, polygons, congruence and vectors
9	B	Angles, polygons, congruence and vectors
10	C	Angles, polygons, congruence and vectors
11	A	Angles, polygons, congruence and vectors
12	B	Angles, polygons, congruence and vectors

Worked solutions

Question 1 GEO-01 · Angles, polygons, congruence and vectors · spec M5.2

The interior angle of a regular polygon is 156° . How many sides does it have?

Answer: F

Exterior angle 24° ; $360/24 = 15$ sides.

Question 2 GEO-02 · Angles, polygons, congruence and vectors · spec M5.2

A polygon has interior angles that sum to 1980° . How many sides does it have?

Answer: E

$$180(n - 2) = 1980 \Rightarrow n - 2 = 11 \Rightarrow n = 13.$$

Question 3 GEO-03 · Angles, polygons, congruence and vectors · spec M5.19

$\mathbf{a} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -3 \\ 4 \end{pmatrix}$. Find $2\mathbf{a} - \mathbf{b}$.

Answer: F

$$\begin{pmatrix} 4 \\ -2 \end{pmatrix} - \begin{pmatrix} -3 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 \\ -6 \end{pmatrix}.$$

Question 4 GEO-05 · Angles, polygons, congruence and vectors · spec M5.5

Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area 20 cm^2 . Find the area of the larger triangle.

Answer: D

$$\text{Area scale factor } \left(\frac{9}{6}\right)^2 = \frac{9}{4}; 20 \times \frac{9}{4} = 45.$$

Question 5 GEO-07 · Angles, polygons, congruence and vectors · spec M5.6

The triangle with vertices (1, 1), (3, 1), (1, 4) is enlarged with scale factor -2 , centre the origin. Which of the following is a vertex of the image?

Answer: C

Multiply each coordinate by -2 : $(-2, -2)$, $(-6, -2)$, $(-2, -8)$.

Question 6 GEO-08 · Angles, polygons, congruence and vectors · spec M5.4

Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?

Answer: A

SSA is not a congruence condition (it is the ambiguous case of the sine rule); the other four are the standard criteria.

Question 7 GEO-09 · Angles, polygons, congruence and vectors · spec M5.2

In the diagram (not shown), AB is parallel to CD . A transversal makes an angle of $(3x + 10)^\circ$ with AB and the co-interior angle with CD is $(2x - 5)^\circ$. Find x .

Answer: E

$$\text{Co-interior angles sum to } 180^\circ: 5x + 5 = 180 \Rightarrow x = 35.$$

Question 8 GEO-10 · Angles, polygons, congruence and vectors · spec M5.3

Which of the following statements is/are true?

- I Every rhombus is a parallelogram.
- II Every rectangle is a rhombus.
- III The diagonals of a kite are perpendicular.

Answer: F

A rhombus has two pairs of parallel sides (I). A rectangle need not have equal sides (II false). The diagonals of a kite cross at right angles (III).

Question 9 GEO-12 · Angles, polygons, congruence and vectors · spec M5.6

A shape is reflected in the line $y = x$ and then rotated 90° anticlockwise about the origin. The point $(3, 1)$ ends up at

Answer: B

Reflection in $y = x$: $(3, 1) \rightarrow (1, 3)$. Rotation 90° anticlockwise: $(x, y) \rightarrow (-y, x)$, so $(1, 3) \rightarrow (-3, 1)$.

Question 10 GEO-13 · Angles, polygons, congruence and vectors · spec M5.2

In triangle ABC , angle A is twice angle B , and angle C is 20° more than angle A . Find angle B .

Answer: C

$$B + 2B + (2B + 20) = 180 \Rightarrow 5B = 160 \Rightarrow B = 32^\circ.$$

Question 11 GEO-14 · Angles, polygons, congruence and vectors · spec M5.5

In triangle ABC , D lies on AB and E lies on AC with DE parallel to BC . $AD = 4$, $DB = 6$ and $DE = 5$. Find BC .

Answer: A

Triangles ADE and ABC are similar with ratio $AD : AB = 4 : 10$. $BC = 5 \times \frac{10}{4} = 12.5$.

Question 12 X2-20 · Angles, polygons, congruence and vectors · spec M5.3

$ABCD$ is a quadrilateral. P : 'the diagonals of $ABCD$ bisect each other'. Q : ' $ABCD$ is a rectangle'. Which one of the following is true?

Answer: B

Every rectangle is a parallelogram, and a parallelogram's diagonals bisect each other: P is necessary. A non-rectangular parallelogram has bisecting diagonals without being a rectangle: not sufficient.